RECEIVER, RADIO R-392( )/URR

1. SCOPE

1.1 This specification covers one type of vehicular communications receiver which provides for reception of radiotelegraph, voice and single channel frequency shift radioteletype signals within a frequency range of 500 kilocycles (kc) to 32 megacycles (mc); designated as Receiver, Radio R-392( )/URR. (See 6.1 and 6.5.)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-571	Solder; Lead Alloy, Tin Lead Alloy, and Tin Alloy; Flux Cored Ribbon and Wire, and Solid Form.

MILITARY

MIL-V-173	Varnish, Moisture-and-Fungus-Resistant, for the Treatment of Communications, Electronic and Associated Electrical Equipment.
MIL-T-713	Twine and Tape, Lacing and Tying (for use in Electrical and Electronic Equipment).
MIL-S-901	Shockproof Equipment, Class HI (High Impact), Shipload.
MIL-I-3158	Insulation Tape, Electrical Glass-Fiber (Resin-Filled): and Cord, Fibrous Glass.
MIL-W-6858	Welding: Aluminum, Magnesium, Non-Hardening Steels or Alloys, and Titanium, Spot, Seam and Stitch.
MIL-I-7798	Insulation Tape, Electrical, Pressure-Sensitive Adhesive, Plastic.
MIL-M-10578	Metal Conditioner and Rust Remover (Phosphoric Acid Base).
MIL-I-11748	Interference Reduction for Electronic Equipment.
MIL-M-13231	Marking of Electronic Items.
MIL-F-14072	Finishes for Ground Signal Equipment.
MIL-R-12454	Radio Set AN/GRC-19( ), Packaging of.
MIL-F-14256	Flux, Soldering, Liquid (Rosin Base).

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STANDARDS

MILITARY

MIL-STD-22	Welded-Joint Designs.
MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-169	Extreme - Temperature Cycle.
MIL-STD-170	Moisture Resistance Test Cycle for Ground Signal Equipment.
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects.

DRAWINGS

SIGNAL CORPS

SC-B-61578	Grounding to Chassis.
SC-DL-144500	Drawing and Data List for Receiver, Radio R-392( )/URR.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies).

2.2 Other publication. - The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28 Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D.C.)

3. REQUIREMENTS

3.1 Description. - Receiver, Radio R-392( )/URR is a ruggedized, immersion-proof receiver which will withstand conditions encountered in mobile (vehicular) military service. The receiver operates at a nominal input voltage of 28.5 volts direct-current (vdc) and is capable of receiving continuous wave (cw), modulated-continuous wave (mcw), voice (am), and frequency shift radio teletypewriter signals. The frequency range is 500 kc to 32 mc in 32 bands. The receiver features an unbalanced antenna input for operation with a 15 foot whip antenna or random length doublet antennas using unbalanced cable. It can be mounted on Mounting MT-851( )/GRC-19 with Radio Transmitter T-195( )/GRC-19, or by itself on Mounting MT-836( )URR. Included as part of the receiver is Case CY-1298( )/URR, which is used for spare parts storage.

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3.2 Preproduction samples. - The contractor shall furnish two preproduction samples of the receiver for approval, provided the invitation for bids and contract require samples. (See 4.3).

3.3 Construction. - Receiver, Radio R-392( )/URR shall be constructed in accordance with the Drawing and Data List SC-DL-144500.

3.4 Parts, materials, and processes. -

3.4.1 Plastic material and parts. - Where not machined, plastic material and parts shall have the original smooth or polished surfaces. Surfaces that have been sawed, cut, punched, or otherwise machined shall be as smooth as practicable in accordance with good manufacturing practice for the intended application.

3.4.2 Controls. - All movable controls shall operate properly without binding or other undue restriction. Controls shall not be assembled in a misaligned condition.

3.4.3 Cleaning. -

3.4.3.1 Parts. - After fabrication, parts shall be cleaned in accordance with good commercial practice, or as specified in an applicable document. Cleaning processes shall have no deleterious effect. Corrosive material shall be removed completely before the parts are mounted on chassis, panels, etc.

3.4.3.2 Units. - After assembly, units shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material. In addition, when necessary, such cleaning shall also be performed before final assembly of the units.

3.4.4 Finish. - The equipment shall be finished in accordance with Specification MIL-F-14072 and the equipment drawings. (See 4.4).

3.4.5 Marking. -

3.4.5.1 Conformance. - Marking shall conform to Specification MIL-M-13231. (See 4.4). Front panel marking shall be Group I as described in that specification.

3.4.5.2 Visibility. - Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.4.5.3 Serial Numbers. - Receiver, Radio R-392( )/URR shall have serial numbers.

3.4.6 Riveting. - Thickness of countersunk metal which accepts the heads of flush rivets shall be not less than the height of the rivet heads. After riveting, the joints shall be tight, the joined parts shall be undamaged, and the rivet heads shall be properly seated and tight against their bearing surfaces.

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3.4.7 Securing of parts. -

3.4.7.1 Mounting items. - Brackets, lugs, flanges, inserts, bolts and other mounting arrangements shall be such as to retain items securely when subjected to specified service conditions. Parts shall be so mounted that loosening, wear of mounting arrangements, or permanent separation of parts will not occur when the items in which they are used are subjected to specified bounce and shock tests. Any permanent deformation resulting from the tests shall not cause degradation of performance below full specification requirements. Parts shall be secured in such manner that failure of a single rivet or screw will not free the part completely.

3.4.7.2 Securing of parts by threaded fasteners. - Nylok or other screws with plastic devices, and Loctite or similar sealants, shall not be subjected to temperatures in excess of 250°Fahrenheit (F) (as during baking of paint). Loctite or similar sealants shall be applied in accordance with the manufacturer's instructions.

3.4.8 Soldering. -

3.4.8.1 Solder. - Solder shall conform to Specification QQ-S-571. For soldering electrical connections, composition Sn60, type AR or S, shall be used for general purposes; and composition Sn62 or Sn63, type AR or S, for special purposes. When type S solder is used for soldering electrical connections, the flux shall conform to Specification MIL-F-14256.

3.4.8.2 Acid or acid salts. - No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning of electrical connections and for tinning or soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use.

3.4.8.3 Process. - There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting actions, and shall not be crystallized, overheated or underheated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in the equipment; flux being spread over a larger area; or damage to the equipment. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

3.4.9 Tropicalization of material. - Material shall be treated as follows:

3.4.9.1 Treating materials. - Treating materials containing a mercury-bearing fungicide shall not be used. The contractor shall determine that the treating material is compatible with the material or surface to be treated. Selection of treating materials shall be such that any increase in flammability of treated material will be held to the practical minimum.

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3.4.9.2 Toxicity. - Treating materials shall cause no skin irritation or other injury to personnel handling the treated material during fabrication, transportation, operation, or maintenance of the equipment, or during use of the finished items when used for purpose intended.

3.4.9.3 Flexibility. - Treatment shall not affect the flexibility of treated materials, to the extent that the equipment may fail to meet specified requirements when subjected to specified service conditions.

3.4.9.4 Statement of treatment. - The contractor shall submit for approval, to the contracting officer, a statement describing in detail the materials to be treated and the treating materials and processes that he proposes to use. (See 6.2).

3.4.10 Welding. - Wherever practicable, welded joints shall conform to Standard MIL-STD-22 and shall be such that grinding on the finished weld will be unnecessary. Spot, stitch, and seam welds shall conform to Specification MIL-W-6858. Spot or other intermittent welds in aluminum or magnesium shall be limited to lightly stressed, non-critical structures. When spot or other intermittent welds are used to hold a part, the number of welds shall be at least two.

3.4.10.1 Cleaning prior to welding. - Surfaces to be welded shall be cleaned in accordance with good commercial practice and shall be free from rust, scale, paint, grease, and other foreign material.

3.4.10.2 Process. - Preheating shall be employed where distortion is likely to result from welding. Welds shall have thorough penetration and good fusion and shall be free from scabs, blisters, abnormal pock marks, cracks, voids, slag inclusions, and other harmful defects. Where undesirable internal stresses are likely to result from welding, welded items shall be stress-relieved. Inert-gas-shielded arc welding shall be used, when practicable, for welding of aluminum, magnesium, or stainless steel.

3.4.10.3 Cleaning after welding. - Welded assemblies shall be cleaned to remove rust, scale, oxidation products, and excess flux by sandblasting, wire brushing, or other suitable means. Prior to painting, steel parts that have been arc welded or acetylene welded shall, in addition, be subjected to vat passivation or a phosphoric acid etch in accordance with Specification MIL-M-10578. Acid used for cleaning shall be completely neutralized and removed.

3.4.11 Wiring and cabling. - Wiring and cabling shall be neat and sturdy.

3.4.11.1 Cabling. - Insulated wires shall be formed into cables as specified in the unit drawings. The cabling of wires shall be effected by tape or twine conforming to Specification MIL-T-713 or tape conforming to Specification MIL-I-7798; however, cord conforming to Specification MIL-I-3158 shall be used where resistance to heat or chemicals is essential. Individual conductors thus secured shall lie essentially parallel; however, this does not prohibit the use of twisted pairs.

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3.4.11.2 Slack. - Wires and cables shall be as short as practical except that sufficient slack shall be provided for the following purposes:

(a) To prevent undue stress on cable forms, wires, and connections, including connections to resiliently supported parts.

(b) To enable parts to be removed and replaced during servicing without disconnecting other parts.

(c) To provide for at least two replacements of the part to which the wire or cable is connected, except for certain rf leads which must be as short as possible for electrical reasons.

(d) To ensure freedom of motion of lugs or terminals normally intended to have some degree of movement (for example, floating contacts on electron tube sockets).

(e) To facilitate field repair of broken or cut wires.

3.4.11.3 Protection. - Wires and cables shall be so placed and protected as to avoid contact, under specified service conditions, with rough or irregular surfaces or sharp edges. Wires shall not be bent sharply where they enter insulation material. Where wires run through holes in metal 1/8 inch in thickness or greater, the wires shall be protected by grommets or bushings or shall have the edges of the holes rounded to a minimum radius of 1/16 inch.

3.4.11.4 Support. - Wire and cable shall be properly supported and secured, to prevent undue stress on the conductors and terminals and undue change in position of the wire or cable (i) during and after subjection of the equipment to specified service conditions or (ii) after service or repair of the equipment in a normal manner. (Note: When shielding on wire or cable is unprotected by an outer insulation, adequate support is necessary to prevent the shielding from coming in contact with exposed terminals or conductors.) Twine or tape shall not be used for securing wire and cable.

3.4.11.5 Clearance. - Clearance between wires or cables, and parts such as electron tubes and resistors, shall be provided to avoid deterioration of the wires or cables because of the heat dissipated by such parts when the equipment is subjected to specified service conditions. Cleance between solder connections or bare conductors, on terminal boards, relays, or other parts, shall be such that no accidental contact can occur between adjacent connections when subjected to specified service-conditions.

3.4.11.6 Splicing. - Wires in a continuous run between two terminals shall not be spliced during the wiring operation.

3.4.11.7 Insulating sleeving. - Insulating sleeving shall be used only as specified.

3.4.11.8 Inductive and capacitive effects. - Wires and cables shall be so located that inductive and capacitive effects, unless used as a design feature, will be the minimum practicable.

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3.4.11.9 Connections. - Before being soldered to terminal lugs or fixed terminals, wires shall be mechanically secured so that the connections are not dependent for strength on solder alone. Where practicable, wires soldered to fixed terminals shall be looped at least one-half turn but no more than three-quarters turn around the terminal before soldering. Bared ends of wire leads to be terminated in solder-type terminals lugs shall be tinned, silver plated, or lead-alloy coated. Electrical connections shall not be made by clamping between a metallic and nonmetallic material. Fraying of textile ends of wires shall be prevented mechanically or by application of varnish conforming to Specification MIL-V-173. No varnish, lacquer, inspection paint, or other coating shall be applied to completed electrical connections.

3.4.11.10 Grounding to chassis. - Grounding to chassis shall conform to Drawing SC-B-61578.

3.4.11.11 Wire and cable shielding. - Wire or cable shielding shall be grounded to the chassis or frame of the equipment. The shielding shall end at a sufficient distance from exposed conductors to insure against shorting or arcing between the conductor and the shielding, and shall be terminated by a crimp-type ferrule or by soldering at the end of the shielding.

3.4.12 Screws, other threaded devices, and related parts. -

3.4.12.1 Form and fit. - Unless otherwise specified, screw threads shall conform to Handbook H28; the Unified Thread Form and Series shall be Class 2A or 2B fit, and the American National Thread Series shall be Class 2 fit.

3.4.12.2 Machine screws. - The screw heads shall be completely seated in the material.

3.4.12.3 Self-tapping screws. - Chips formed in through holes by thread-cutting screws shall be removed.

3.4.12.4 Thread engagement and extension. - The length of thread engagement for tapped holes in steel or hard brass shall at least equal the nominal thickness of corresponding machine-screw nuts in Sizes No. 12 and smaller and semifinished hexagon regular nuts in sizes 1/4 inch and larger, as shown in Handbook H28. Thread engagement for tapped holes in aluminum, magnesium, or other soft material shall be at least 50 percent greater than the thread engagement specified for steel or hard brass. Where permitted by design of the equipment, screws and bolts shall extend at least 1-1/2 threads beyond the nut, but not more than 1-1/2 threads plus 1/8 inch for screws up to 1 inch in length and 1-1/2 threads plus 1/4 inch for screws over 1 inch in length.

3.4.13 Electron tubes and crystal units. - The equipment shall be furnished with the same electron tubes and crystal units used when the set was inspected for acceptance. Removable items shall remain in the respective sockets occupied during inspection of the equipment. Specified performance of the equipment shall be obtained without selection of tubes.

3.5 Interchangeability. - Like units, assemblies, subassemblies, and replaceable parts shall be physically and functionally interchangeable, without modification of

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such items or of the equipment. (See 4.20). Individual items shall not be hand-picked for fit or performance; however, matched pairs or sets, when permitted, may be interchangeable as such. Reliance shall not be placed on any unspecified dimension, rating, characteristics, etc.

3.6 Air-seal test, provision for. - The equipment enclosure shall provide means for enabling performance of the air-seal test. (See 4.18).

3.7 Suppression of radio interference (see 4.21). - The equipment shall conform to Specification MIL-I-11748 and the Class I limits specified therein.

3.8 Service conditions. - The equipment shall meet the following service conditions--where a test is referenced, meeting the test shall be considered as compliance with the requirement.

3.8.1 Operation. - Continuous use for a period of 1 year, under the worldwide environmental conditions specified in 3.8.2 through 3.8.7, with a duty cycle of 23 hours on and 1 hour off and with no more than normal maintenance and replacement of parts. Where the contractor is required to make a selection of parts, materials, processes, construction methods, etc., he shall be guided by this requirement. Approval of the preproduction sample (see 3.2) shall be considered as compliance with this requirement.

3.8.2 Temperature (see 4.9). -

(a) Operating: Ambient temperature in the range of +150°F to -40°F. (The 150°F temperature includes effect of sun-load). Exposure at the high temperature extreme not to exceed 4 hours, and at the low temperature extreme not to exceed 72 hours, at any one time.

(b) Nonoperating: Exposure in the range of +160°F to -80°F; exposure at the high temperature extreme not to exceed 4 hours, and at the low temperature extreme not to exceed 24 hours, at any one time.

3.8.3 Relative humidity (see 4.10). - Up to 97 percent relative humidity for 20 hours; and exposure at 100 percent relative humidity, with condensation, for 4 hours.

3.8.4 Elevation (see 4.11). -

(a) Operating: Up to 10,000 feet above sea level.

(b) Nonoperating: Up to 25,000 feet above sea level.

3.8.5 Immersion (see 4.15). - Three feet of water for 2 hours.

3.8.6 Orientation. -

(a) Operating (see 4.12). - Any orientation up to 20 degrees from normal operating position (that is: forward, backward, left or right).

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(b) Nonoperating: Storage in any position for a period of two years. Where the contractor is required to make a selection of parts, materials, processes, construction methods, etc. he shall be guided by this requirement. Approval of the preproduction sample (see 3.2) shall be considered as compliance with this requirement.

3.8.7 Vibration, bounce, and shock. - As described in 3.9.

3.9 Vibration, bounce, and shock. -

3.9.1 Vibration, Internal. - The amplitude of vibration of any part, subassembly, or structural member of the equipment shall not exceed twice the amplitude of the vibration applied to the equipment at any frequency between 10 and 55 cycles per second. (See 4.16).

3.9.2 Bounce and shock. - The receiver shall be capable of meeting the requirements of Table I.

Table I. - Bounce and shock

Paragraph Heading	Paragraph Number	Performance after test
Bounce	4.17	Specified performance. (Note A). No physical damage.
Shock; ballistic	4.13	Operable. (Note B). Any physical damage shall be minor only.
Shock; benchhandling	4.14	Specified performance. (Note B). No physical damage.

Note A: Not more than one error in 10 consecutive lines of teletype copy.

Note B: The receiver shall meet specified performance for the following tests before and after the shock test: overall gain (4.8.11), IF output (4.8.19), and audio output (4.8.14).

3.10 Electrical requirements (see 4.8). - The receiver shall be capable of performing all its functions with input voltage from 22 to 30 vdc. Reduced performance is permitted with input potentials below 24 volts.

3.10.1 Frequency vs supply voltage (see 4.8.1). - Frequency variation with a change in supply voltage from 22 to 24 volts dc shall not exceed 200 cps, and with a change in supply voltage from 24 to 30 volts dc shall not exceed 125 cps.

3.10.2 Dial lock frequency error (see 4.8.2). - Frequency shift due to action of the dial lock shall not exceed 50 cps.

3.10.3 BFO operation (see 4.8.3). - The BFO pitch when in the counterclockwise (ccw) and clockwise (cw) position shall be not less than 3 kc and not more than 4.5 kc.

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3.10.4 IF and image rejection (see 4.8.4. - The if and image rejection ratio shall be not less than 80 db, except that the image rejection of the 2-3 mc if at a dial frequency of 31.980 mc shall be not less than 60 db.

3.10.5 IF gain vs bandwidth (see 4.8.5). - The difference in gain between any two if bandwidth settings shall not exceed 6 db.

3.10.6 Spurious rejection (see 4.8.6). - The spurious rejection ratio shall be not less than 55 db, except that the spurious rejection at dial frequencies of 6.020 mc and 24.100 mc shall be not less than 40 db, and 2.480 mc shall be not less than 50 db.

3.10.7 Internal signals (see 4.8.7). - Over the tuning range of the receiver, internally generated signals shall be not greater than the equivalent of a 1.0 microvolt signal injected into the antenna through a 56 uufd capacitor dummy antenna, except that internal signals at 2.727 and 3.227 mc shall be not greater than the equivalent of a 3.0 microvolt signal.

3.10.8 Selectivity (see 4.8.8). - Overall selectivity for each position of the if bandwidth switch shall be as follows:

Input (uv)	Attenuation db down	2 Kc limit	4 Kc limit	8 Kc limit
7.07	3.0 +			NLT 3.2 kc
	-			NLT 3.2 kc
	total			NLT 7.0 kc
10.0	6.0 +	NLT 0.8 kc	NLT 1.5 kc	
	-	NLT 0.8 kc	NLT 1.5 kc	
	total	NLT 1.9-2.6 kc	NLT 3.2-4.6 kc	
5K	60.0 total	NMT 8.5 kc	NMT 8.5 kc	NMT 22.0 kc

NMT - Not more than
NLT - Not less than

Peak to valley ratio across the pass band shall be not more than 2 db.

3.10.9 AM and CW sensitivity (see 4.8.9 and 4.8.10). - The sensitivity of the receiver at a signal plus noise to noise power output ratio of ten to one, with a power output of ten milliwatts and with the if bandwidth switch in the 8 kc position, shall be as specified below.

Frequency range	AM	CW
.50 to .99	9.0 uv	3.0 uv
1.00 to 2.99	7.0 uv	3.0 uv
3.00 to 3.99	6.0 uv	2.0 uv
4.00 to 5.99	7.0 uv	2.0 uv
6.00 to 31.99	5.0 uv	2.0 uv

Limits: S+S/N shall not be less than 10 db for both CW and AM sensitivity.

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3.10.10 Overall gain (see 4.8.11). - The overall gain of the receiver shall be such that a diode load voltage of -2.5 volts is produced with a signal input of not more than 4 microvolts (uv) and not less than 1 microvolt.

3.10.11 AGC characteristics (see 4.8.12). - The automatic gain control shall be such that the receiver power output shall meet the below listed limits:

Input	Output change (db)	Carrier level meter reading
2.5 uv	NMT 3 db	NMT 15
5 uv	5 mw ref.	NMT 25
1000 uv	NMT 4 db	NLT 25
100k uv	NMT 8 db	NLT 50
1 volt	NMT 16 db	NLT 60

3.10.12 Squelch threshold (see 4.8.13). - Not more than 3.5 volts developed at the diode load by the input signal shall be required to open the squelch circuit and not less than 0.8 volts developed at the diode load by the input signal shall be required to close the squelch circuit.

3.10.13 Audio output (see 4.8.14). - When measured as specified in 4.8.14, audio output shall be not less than 1,000 milliwatts.

3.10.14 Distortion (see 4.8.15). - Distortion shall be no more than 12 percent with one watt output; when measured in accordance with 4.8.15.

3.10.15 Overall audio response (see 4.8.16). - With 1000 cps as a reference level, the overall audio response shall be flat within ± 1 db to 300 cps and within +1 to -3 db to 3500 cps.

3.10.16 Hum level (see 4.8.17). - With one volt of 400 cps ac superimposed on the 28.5 volt dc power source, the level of the 400 cps tone at the audio output shall be not less than 30 db below a one watt reference level.

3.10.17 Noise limiter (see 4.8.18). - The modulation level at which the noise limiter starts to clip shall be no more than 55% and no less than 35%.

3.10.18 IF output characteristics (see 4.8.19). - With an input of 3 microvolts, the if output voltage shall be not less than 10 millivolts for all settings of the if band-switch and with the AGC switch in either the ON or the OFF position. With a signal input of 5 microvolts, the if section shall have a signal-plus-noise to noise ratio of not less than 10 to 1.

3.10.19 Break-in relays (see 4.8.20). - The antenna grounding relay attenuation shall be not less than 8 db when measured at 30.05 mc. The audio disabling relay shall function properly when checked as required in 4.8.20.

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3.10.20 Tuning knob torque (see 4.8.21). - The kilocycle and megacycle tuning knob shafts shall be capable of withstanding repeated torque of 72 inch pounds in each direction without permanent deformation.

3.10.21 Backlash (see 4.8.22). - Backlash between the kilocycles knob and the frequency indicating counter shall not exceed 50 cps. Backlash between the kilocycles knobs and the variable frequency oscillator shall not exceed 50 cps.

3.10.22 Megacycle dial calibration error (see 4.8.23). - Frequency error between any two 1 mc tuning ranges, due to error in the crystal oscillator injection frequencies shall not exceed 3 kc.

3.10.23 Frequency identification accuracy (see 4.8.24). - When the indicating counter has been adjusted at the nearest 100 kc calibration point, the receiver shall tune to within 300 cps of the desired frequency.

3.10.24 Zero-adjust mechanism (see 4.8.25). - The dial zero adjust mechanism shall provide ± 6 kc correction.

3.10.25 100 kc Crystal calibrator. -

3.10.25.1 Calibrator output (see 4.8.26.1). - The crystal calibrator shall furnish 100 kc calibration signals across the receiver tuning range. No more than six calibrator signals across the receiver tuning range shall fail to energize the carrier level meter. The calibrator signal level shall be sufficient to provide at least a signal-plus-noise to noise ratio of 10:1.

3.10.25.2 Crystal calibrator cycling (see 4.8.26.2). - Frequency changes in the output of the 100 kc crystal calibrator due to temperature changes during crystal oven cycling shall not exceed 100 cps when the receiver is tuned to 30.0 mc.

3.10.26 Tracking of tuned circuits (see 4.8.27). - Mistracking in the antenna circuit with the antenna trimmer set at 0, shall not exceed 1 db. Mistracking in the three succeeding rf circuits shall not exceed a total of 3 db. Mistracking in the three 9-18 mc variable if tuned circuits shall not exceed a total of 3 db. Mistracking in the three 2-3 mc variable if tuned circuits shall not exceed a total of 3 db.

3.10.27 Desensitization (see 4.8.28). - Desensitization, when measured in accordance with 4.7.39, shall comply with the following:

Level of Undesired Signal	Maximum separation between desired and undesired signals for 3 db depression of desired signal
0.1 volt	50 kc
1.0 volt	100 kc
10.0 volts	200 kc

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3.10.28 Cross modulation (see 4.8.29). - Cross modulation response shall comply with the following limits when tested in accordance with 4.8.29:

Level of Undesired Signal (volts)	Desired Signal (mc)	Maximum separation between Desired and Undesired Signals (kc)
0.01	3.0	50
0.1	3.0	150
1.0	3.0	400
10.0	3.0	1000
0.01	6.0	100
0.1	6.0	200
1.0	6.0	600
10.0	6.0	1600
0.01	12.0	100
0.1	12.0	200
1.0	12.0	1000
10.0	12.0	3500

3.10.29 AGC time constant (see 4.8.30). - With the BFO switch in the ON position, the AGC time constant (discharge time) shall be 0.5 ± 0.1 seconds. With the BFO switch in the OFF position, the AGC time constant shall be 0.05 ± 0.01 seconds.

3.10.30 Carrier level meter (see 4.8.31). - As the signal input to the receiver is increased in 20 db steps, the reading on the carrier level meter shall increase by 20 db ± 6 db.

3.10.31 Blocking (see 4.8.32). - With AGC on, the receiver shall not block with signal inputs up to .2 volt. The input required to cause blocking shall be determined by noting the point at which the audio output begins to decrease with increasing signal input.

3.10.32 Variable frequency oscillator (VFO) stability vs temperature (see 4.8.33). - Frequency stability vs temperature of 2.455 to 3.455 mc VFO shall be such that anywhere within the tuning range, the frequency shall change NMT 300 cycles as the temperature is changed from +68°F to 104°F (+20°C to 40°C), and not more than 700 cycles as the temperature is changed from +50°F to +140°F (+10°C to +60°C).

3.10.33 Beat frequency oscillator (BFO) stability vs temperature (see 4.8.34). - The 455 kc BFO shall be temperature compensated so that the point of least frequency change with temperature change, occurs when the ambient temperature around the receiver is in the range of 68°F to 104°F (20°C to 40°C). Frequency stability vs temperature in steps of every 10 degrees shall be within the following limits: Frequency variation at 32°F from that at 86°F not to exceed 120 cps maximum, at 140°F from that at 86°F not to exceed 80 cps maximum, at 176°F from that at 86°F not to exceed 105 cps maximum.

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3.10.34 BFO leakage (see 4.8.35). - With the BFO tuned for maximum leakage, the BFO leakage at the if output shall not exceed 4000 microvolts.

3.10.35 20-hour frequency stability (see 4.8.36). - The overall frequency change, as measured at the audio output, starting three minutes after power is applied, shall not exceed 900 cps during the first two hours and shall not exceed 500 cps from the second to the 20th hour. Any sudden frequency variations in excess of 10 cycles per second shall not exceed 25 cps. No more than three such variations shall occur during the test period.

3.10.36 Operational inspection. - The operational test specified in 4.22 shall be performed to determine that the receiver is operable, prior to packaging and at such other times as required.

3.11 Preconditioning. - The equipment shall be capable of meeting the requirements herein, without subsequent processing, after subjection to the bounce preconditioning of 4.6 (also see 4.5).

3.12 Technical literature and running spare parts. - Technical literature, tools and running spare parts shall be furnished if specified in the contract. Running spare parts shall be identical to corresponding parts in the equipment furnished on the order. (See 6.2).

3.13 Workmanship. - The equipment shall be manufactured and assembled in accordance with the applicable portions of the following paragraphs herein:

3.4.1	Plastic material and parts.	3.4.6	Riveting
3.4.2	Controls	3.4.7	Securing of parts.
3.4.3	Cleaning.	3.4.8	Soldering.
3.4.4	Finish.	3.4.9	Tropicalization of material.
3.4.5	Marking.		

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. - The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. - Inspection shall be classified as follows:

(a) Preproduction inspection (does not include preparation for delivery). (See 4.3).

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(b) Acceptance inspection.

(1) Acceptance inspection of equipment before preparation for delivery. (See 4.4 and 4.5).

(2) Acceptance inspection of preparation for delivery. (See 4.23).

4.3 Preproduction inspection. - This inspection will be performed by the Government unless otherwise specified in the contract. It shall consist of the preproduction inspection specified in table II, the inspection specified in the subsidiary documents covering the items listed in 4.4, and the inspection specified for group A, group B, and group C (see tables III, IV, and V respectively). The preproduction inspection will normally be performed in this order: (1) vibration, (2) bounce, (3) shock, bench-handling, (4) shock, ballistic, and (5) immersion; other preproduction inspection may precede, follow, or be interspersed between the foregoing.

Table II - Preproduction inspection

Inspection (For additional preproduction inspection see 4.3)	Req para.	Insp. para.
Suppression of radio interference	3.7	4.21
Temperature	3.8.2	4.9
Moisture-resistance	3.8.3	4.10
Elevation	3.8.4	4.11
Orientation	3.8.6	4.12
Shock test; ballistic	3.9.2	4.13
Shock test; bench-handling	3.9.2	4.14
Vibration; resonance search	3.9.1	4.16
Bounce	3.9.3	4.17

4.4 Acceptance inspection covered by subsidiary documents. - The following shall be inspected under the applicable subsidiary documents as part of the acceptance inspection before preparation for delivery:

Item	Where required
Finish	3.4.4
Marking	3.4.5

4.5 Acceptance inspection of equipment before preparation for delivery. - The contractor, to demonstrate compliance with specified requirements, shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.3.2. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection

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procedures and examine the contractor's inspection records. In addition the Government-- at its discretion -- may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. (See 6.7). Test equipment for Government verification inspection shall be made available by the contractor. Each unit which will be subjected to group A, group B, or group C inspection shall be preconditioned after final assembly. (See 3.11).

4.5.1 Group A inspection. - This inspection, including sampling, shall conform to table III and the ordinary inspection procedures of Standard MIL-STD-105. Group A inspection shall be performed in any order which is satisfactory to the Government, except that the air-seal test (4.18) shall be next to last and the operational inspection (4.22) shall be last. (See 6.5).

Table III - Group A inspection

Inspection	Req. Para.	Insp. Para.	AQL Major	AQL Minor
Visual and mechanical	3.13	4.19		
RF and gear train sub-chassis			2.5 dphu*	10.0 dphu
Crystal oscillator sub-chassis			1.0 dphu	4.0 dphu
Calibration oscillator sub-chassis			1.0 dphu	4.0 dphu
IF sub-chassis			1.5 dphu	6.5 dphu
AF sub-chassis			2.5 dphu	10.0 dphu
VFO mixer sub-chassis			1.0 dphu	4.0 dphu
BFO assembly			1.0 dphu	4.0 dphu
Lower deck assembly				
Crystal calibrator sub-chassis		)		
IF sub-chassis		)	2.5 dphu	10.0 dphu
BFO assembly		)		
Front panel assembly			1.0 dphu	4.0 dphu
Audio module			1.0 dphu	4.0 dphu
Terminal board assemblies			1.0 dphu	4.0 dphu
Completed assembly			4.0 dphu	15.0 dphu
Electrical				
Frequency vs. supply voltage	3.10.1	4.8.1)		
Dial lock frequency error	3.10.2	4.8.2)		
Internal signals	3.10.7	4.8.7)		
Selectivity	3.10.8	4.8.8)		
AM sensitivity	3.10.9	4.8.9)		
CW sensitivity	3.10.9	4.8.10)	4.0 dphu	
Overall gain	3.10.10	4.8.11)	for the group	
AGC characteristics	3.10.11	4.8.12)		
Squelch threshold	3.10.12	4.8.13)		
Audio output	3.10.13	4.8.14)		
Distortion	3.10.14	4.8.15)		
IF output	3.10.18	4.8.19)		
Megacycle dial calibration error	3.10.22	4.8.23)		
Air seal	3.6	4.18	1%	
Operational test	3.10.6	4.22	1%	

*Dphu - defects per hundred units.

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4.5.2 Group B inspection. - This inspection, including sampling, shall conform to table III and to the special procedures for small-sample inspection of Standard MIL-STD-105. The reduced inspection procedure shall be R-1. Group B inspection shall normally be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection.

4.5.2.1 Group B sampling plans. - The group B sampling plans, for the AQL's listed in table IV, shall be as follows:

AQL	Inspection level for normal and tightened inspection	Inspection level for reduced inspection
6.5%	L-7	L-5

4.5.2.2 Order of inspection within group B. - Group B inspection shall be performed in any order which is satisfactory to the Government.

4.5.2.3 Procedure in case of failure. - When an inspection lot fails, the contractor shall immediately investigate the cause of failure and take corrective action to assure that subsequent lots do not contain the same defect or defects.

Table IV - Group B inspection

Inspection	Req para	Insp para	AQL (see 4.5.2.1)
Interchangeability	3.5	4.20	6.5%
BFO operation	3.10.3	4.8.3	6.5% for the group combined
IF and image rejection	3.10.4	4.8.4	
IF and gain vs. bandwidth	3.10.5	4.8.5	
Spurious rejection	3.10.6	4.8.6	
Overall audio response	3.10.15	4.8.16	
Hum level	3.10.16	4.8.17	
Noise limiter	3.10.17	4.8.18	
Break-in relays	3.10.19	4.8.20	
Backlash	3.10.21	4.8.22	
Frequency identification accuracy	3.10.23	4.8.24	
Zero adjust mechanism	3.10.24	4.8.25	
100 kc crystal calibration	3.10.25	4.8.26	
Calibrator output	3.10.25.1	4.8.26.1	
Crystal calibrator cycling	3.10.25.2	4.8.26.2	
Desensitization	3.10.27	4.8.28	
Carrier level meter	3.10.30	4.8.31	
Blocking	3.10.31	4.8.32	
BFO leakage at IF output	3.10.34	4.8.35	
20 hour frequency stability	3.10.35	4.8.36	6.5%

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4.5.3 Group C inspection. - This inspection shall be as listed in Table V, and shall normally be performed on sample units that have been subjected to and met group A and group B inspection.

Table V - Group C inspection

Inspection	Rqt Para.	Insp Para.
Tuning knob torque	3.10.20	4.8.21
Tracking of tuned circuits	3.10.26	4.8.27
Cross modulation	3.10.28	4.8.29
AGC time constant	3.10.29	4.8.30
VFO stability vs temperature	3.10.32	4.8.33
BFO stability vs temperature	3.10.33	4.8.34
Immersion	3.8.5	4.15

4.5.3.1 Sampling for inspection of equipment. - For the first group C inspection, two receivers from the first months production, or the first 50, whichever is less, shall be selected. For subsequent group C inspections, two from each successive 100, or two per month if less than 100 are produced monthly, shall be selected.

4.5.3.2 Noncompliance. - If a sample unit fails group C inspection, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken on the process and all units of product which were manufactured with the same conditions, materials, processes, etc. If the Government inspector does not consider that the corrective action will enable the product to meet specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer. (See 6.4)

4.6 Bounce preconditioning. - The unit, shall be placed in its normal operation position on the table of the Package Tester as made by the L.A.B. Corporation, Skaneateles, New York, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32-inch-thick strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. The unit shall be subjected to this preconditioning for 1 minute. After bounce preconditioning, the unit shall not be repaired, alined, cleaned, or otherwise changed prior to subjection to acceptance inspection.

4.7 Standard conditions. - Unless otherwise specified, the electrical tests shall be performed under the following conditions:

(a) Supply voltage shall be 28.5 ± .5 vdc.

(b) A standard audio load or headphones of 600 ohms shall be used.

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(c) A 56 uuf capacitor dummy antenna shall be used to couple the receiver to the signal generator, and the signal generator output shall be 30 percent modulated with 400 cps.

(d) The receiver controls shall be set as follows: ANT TRIM set for peak diode load voltage at each dial frequency fed into the receiver, function switch set to NORMAL, BFO switch to OFF, AGC switch to AGC ON, RF GAIN fully clockwise, AF GAIN fully clockwise, and BANDWIDTH to 8 kc.

(e) Normal room ambient conditions of temperature and humidity.

4.8 Electrical tests.-

4.8.1 Frequency vs. supply voltage (See 3.10.1).- With a secondary frequency standard connected to the receiver ANT post, attach a frequency counter across the audio load. Turn the B.F.O. to ON, tune receiver to 25 mc, obtaining a convenient beat note (1000 cps) on the frequency counter. Allow the system to stabilize for a few minutes at 24 volts dc input, then record frequency counter indication. Adjust power supply to 30 volts input and record the change in the beat note. Adjust to 22 vdc input and record frequency change.

4.8.2 Dial lock frequency error (See 3.10.2).- With a secondary frequency standard connected to the receiver ANT post, set the controls as in 4.7, but with the BFO to ON. Obtain a convenient beat note (1000 cps) on a frequency counter at 25 mc. Tighten DIAL LOCK and record change in beat note as indicated by the frequency counter.

4.8.3 B.F.O. operation (See 3.10.3).- Set BFO PITCH at 0 and adjust receiver KILOCYCLES control to obtain zero beat with the calibration oscillator. Turn BFO PITCH clockwise, then counterclockwise and record maximum indication on a frequency counter for each direction.

4.8.4 IF and image rejection (See 3.10.4).- With a signal generator, Measurements Corporation, Model 32, or equal, connected under standard conditions, tune the coarse control to 1 mc and calibrate the generator output. A dc VTVM shall be connected to J-615 on the receiver (diode load voltage) and the receiver tuned to 7.960 mc. With the receiver controls set as in 4.7, but with AGC off, adjust the HF attenuator on the generator to 3 microvolts and tune the HF Fine control to obtain maximum indication on the VTVM (maximum diode load voltage). Connect the 600 ohm audio load to the receiver and adjust the receiver AF GAIN to obtain 10 millwatts (mw) on the audio load meter. Tune the generator to an if frequency of 17.960 mc and adjust the fine frequency control to obtain maximum diode load voltage. Adjust the HF attenuator on the signal generator to obtain 10 mw audio output and record the attenuator reading. Repeat for a dial frequency of 31.980 mc.

4.8.5 IF gain vs bandwidth (see 3.10.5).- Adjust the signal generator for peak response with the receiver BANDWIDTH switch in the 2 kc position, and adjust the

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generator output level to produce -2.5 volts across the diode load terminals. Without further adjustment, the IF BANDWIDTH switch shall be rotated to the 4 kc and 8 kc positions. The change in diode load voltage, in db, shall be recorded for each setting.

4.8.6 Spurious rejection (See 3.10.6). - The test setup for spurious rejection shall be the same as 4.8.4, except that a receiver dial and signal generator frequency of 2.480 mc shall be used. Tune the signal generator to 2.520 mc, advance the signal generator output until a diode load voltage above noise voltage is indicated and peak with the Fine frequency control on the signal generator. Adjust the H.F. attenuator on the signal generator to again obtain 10 mw audio output and record the attenuator reading. Use a crystal diode modulator while performing this test. Repeat the above test for the following dial frequencies: 6.020 mc, 8.230 mc, 9.020 mc, and 24.100 mc.

4.8.7 Internal signals (See 3.10.7). - With receiver controls set as in 4.7, but with BFO to ON, AGC OFF, and with no connection to ANT post, tune receiver to 2.927 and search for a signal. When signal is found, turn BFO switch to OFF and tune receiver for maximum diode load voltage as read on a DC VTVM connected to J-615 on the receiver. Record this diode load voltage and connect signal generator (using no modulation) through 56 uuf dummy antenna and tune the generator to 2.712 mc. Tune the receiver to 2.712 and adjust H.F. fine frequency dial on the signal generator to obtain maximum diode load voltage. Adjust signal generator attenuator to obtain same diode voltage as recorded above. Record the attenuator reading in microvolts as the strength of the internal signal. Repeat the above for frequencies of 2.727 and 3.227 mc. The requirements of 3.10.7 shall be met.

4.8.8 Selectivity (See 3.10.8). - Set receiver controls as in 4.7 except AGC shall be off. Connect a VTVM to diode load J-615 and a frequency counter to audio output of receiver. Tune the receiver 4.40 mc. Connect a signal generator to the ANT post on the receiver through the 56 uuf dummy antenna and tune the signal generator to 4.4 mc. Turn BFO to ON and BFO PITCH to 0. Adjust the attenuator on the signal generator to 5 microvolts. Then adjust signal generator HF fine control to obtain a zero beat. Turn BFO OFF. Set the RF Gain control on the receiver to obtain a convenient diode load voltage of approximately 2.5 volts. Note this diode load voltage. Adjust the attenuator on the signal generator to 7.07 microvolts and tune the HF fine frequency control to one side until the same diode load voltage as noted previously is obtained. Turn BFO on and note frequency indicated on the frequency counter. Turn BFO OFF and adjust fine frequency to other side until the same diode load voltage is obtained. Turn BFO ON and note counter indication. Selectivity curve should be symmetrical (each frequency reading should be approximately equal). Tune the signal generator across the pass band and record the db difference between the peak and the valley as read on a VTVM connected to the J-615. Repeat above steps for 4 kc and 2 kc bandwidths using the signal input voltages indicated in 3.10.8.

4.8.9 AM sensitivity (See 3.10.9). - With the receiver controls set as in 4.7, but with the AGC OFF, connect the signal generator through a Crystal Diode modulator to receiver ANT connector using the dummy antenna. Connect the signal generator LF OUT to the MOD IN connector on a crystal modulator use a "T" connector on the MOD IN so that an ac VTVM may also be connected at this point. Connect a dc meter to the

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BIAS + and - terminals on the crystal modulator and note that the proper bias exists for that modulator, if not adjust bias control for proper bias. Tune the receiver 500 kc. Set the signal generator attenuator (after calibrating it) to 15 microvolts, and adjust the HF coarse and HF Range to obtain .500 megacycles output. Tune the signal generator HF fine to obtain maximum receiver diode load voltage as read on the dc VTVM connected to J-615. Set the signal generator attenuator for an imput of 9 uv. Adjust the signal generator to obtain 400 cps modulation. Then adjust the L. F. gain to obtain 3 volts as read on the ac VTVM connected to the "T" at MOD IN on the crystal modulator. Connect and audio load of 600 ohms impedance. Adjust the receiver AF GAIN to obtain 10 milliwatts audio output as read on an audio load meter, and remove the modulation. Record the db drop in audio output. Repeat the above steps for following frequencies: 0.50, 0.75, 0.99, 1.00, 1.50, 1.99, 2.00, 3.00, 4.00, 6.00, 7.99, 8.00, 12.00, 15.99, 18.00, 24.00, and 31.99 mc at the inputs indicated in 3.10.9.

4.8.10 CW sensitivity (see 3.10.9).- Set up receiver the same as for AM sensitivity test, with receiver tuned to 750 kc. Turn bandwidth control to 4 kc position. Set attenuator on signal generator to 3.0 microvolts on a signal generator frequency of .750 megacycles. Turn toggle switch to OFF (down position), tune HF fine frequency dial on the signal generator to obtain a maximum diode load voltage on the receiver. Turn receiver BFO to ON and adjust BFO PITCH control to obtain approximately a 1000 cps beat and adjust AF GAIN to obtain 10 milliwatts output on the 600 ohm load. Turn the signal generator dial approximately 20 kc off the frequency and record the db drop in audio output. Repeat above test for the following frequencies 0.75, 1.50, 3.00, 6.00, 12.00, and 24.00 mc at the input indicated in 3.10.9.

4.8.11 Overall gain (see 3.10.10).- With receiver controls set as in 4.7, but with AGC OFF, connect signal generator to receiver through dummy antenna, the receiver dial set to 500 kc, connect a VTVM to J-615 and adjust signal generator HF coarse and HF fine to obtain maximum receiver diode load voltage at .5 mc. Adjust signal generator attenuator to obtain 2.5 volts diode load and record attenuator setting. Repeat above test for all frequencies from 1.5 to 31.5 mc in 1 mc steps.

4.8.12 AGC characteristics (see 3.10.11).- With receiver controls set as in 4.7, connect a signal generator to the receiver ANT post through the 56 uuf dummy antenna. Adjust the attenuator in 5 microvolts after calibration. Tune the receiver to 1 mc. Adjust the signal generator HF coarse to HF fine range to obtain 1 mc output. Zero beat with BFO ON, and then turn BFO OFF. Adjust LF dial on the signal generator to 400 cps. Adjust the LF gain to obtain 30% modulation. Adjust the AF gain on the receiver to obtain 5 mw output as read on the audio load meter connected to the receiver audio output. Connect a distortion analyzer TS-723 ( )/U, or equal, input terminals to the audio load and set distortion analyzer function switch to set level, the meter switch to 10% (0 db) and adjust the input control to obtain 0 db. Decrease the attenuator on the signal generator to 2.5 microvolts and record the decrease in output read directly in db on the distortion analyzer meter. Record CARRIER level meter reading. Record the increase in output for all other input voltages listed in 3.10.11, also record the receiver carrier level meter indication. (Note: The generator must be zero beat as signal level is

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increased). Connect the VTVM to J-614 (AGC voltage) and adjust the signal generator to 100 k microvolts. Turn BFO ON and remove connector from ANT connector, removing signal input. Turn BFO OFF, connect and then disconnect input signal. Verify that decay time (time required for meter indication to drop to a low value) is no longer with BFO ON than with BFO OFF. The above readings and recordings shall meet the requirements of 3.10.11.

4.8.13 Squelch threshold (see 3.10.12). - With the receiver controls set in 4.7 and a dc VTVM connected to J-615 on the receiver, the function switch to SQ, connect the signal generator to the ANT post through the 56 uuf dummy antenna and adjust the signal generator attenuator to 50 microvolts. Set signal generator HF coarse and HF fine range to any convenient frequency and LF dial to obtain 400 cps and adjust LF gain to 30%. Set receiver MEGACYCLES and KILOCYCLES to the above convenient frequency. Tune the signal generator HF fine to obtain maximum diode load voltage as indicated by a dc VTVM connected to J-615 on the receiver. Turn rf gain control on receiver fully counterclockwise, then slowly advance rf gain clockwise and note diode load voltage at the point where the squelch opens. Also note where audio signal is heard from speaker connected to receiver audio output, and light on panel marked SQUELCH comes on. Then slowly turn rf gain control counterclockwise and note the diode load voltage at which the squelch closes, no audio is heard, and squelch light is extinguished.

4.8.14 Audio output test (see 3.10.13). - With the controls set as in 4.7, adjust the MEGACYCLES dial to 01 and the KILOCYCLES dial to 000- Apply a 10- microvolt, 1-megacycle signal, modulated 30 percent at 400 cycles per second from a signal generator through a 56- micromicrofarad capacitor to the ANT terminal on the receiver. Connect a dc vtvm to J-615 on the receiver and tune the signal generator for maximum diode load voltage. Connect a General Radio Type 583 power output meter, or equal, adjusted for a 600- ohm load, to the receiver audio output terminals. Record the audio output. Remove the modulation from the signal generator, change its rf output to 4 microvolts, turn the BFO switch to ON, and again record the audio output. Both recorded outputs shall meet the requirements of 3.10.13.

4.8.15 Distortion (see 3.10.14). - With the receiver controls, set as in 4.7, set the MEGACYCLES control to 01 and the KILOCYCLES dial to 500. Connect a signal generator set at 1.5 mc and adjust the attenuator to 1000 microvolts output. Tune HF fine for maximum diode load voltage. Adjust LF dial on the signal generator to 400 cps. Then adjust LF gain on signal generator to obtain 30% modulation. Adjust the receiver AF gain to obtain 1 watt audio output on the audio load. Connect a distortion analyzer's input terminals to an audio load and turn the function switch on the analyzer to SET LEVEL, the meter switch to 100% and adjust the input control to obtain full scale reading. Change the function switch to the DISTORTION position and adjust the frequency dial to 400 cps and range switch to XI. Tune frequency dial and balance control to obtain a minimum meter indication. Switch to the 30% meter position and again adjust the frequency dial and the balance control for a minimum. Record the meter reading as percent distortion.

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4.8.16 Overall audio response (see 3.10.15).- With the receiver controls set as in 4.7, and the frequency of the receiver and signal generator at 1.5 mc, turn on the BFO and set BFO PITCH to 0. Adjust the signal generator frequency to obtain zero beat. Turn off BFO. With the attenuator on the signal generator set at 1000 microvolts, change the LF dial to 1000 cps and adjust LF gain for 30% modulation. Adjust the AF gain to obtain 10 milliwatts on the audio load. With a distortion analyzer connected to the audio load, turn distortion analyzer function switch to Set Level and the meter switch to 100% (+20 db). Adjust input control to obtain 0 db on the meter. Change LF dial on the signal generator to 300 cps keeping the percent modulation at 30%. Note and record the db above (+) or below (-) the 0 db reference level as read on the distortion analyzer meter. Repeat above steps at 600, 1000, 1800, 2500, and 3500 cps.

4.8.17 Hum level (see 3.10.16).- With the receiver controls set as in 4.7, and the signal generator tuned for maximum receiver diode load voltage at 1.5 mc, turn the LF dial on the signal generator to 400 cps and the LF gain to obtain 30% modulation. Set the attenuator at 10 K uv output. Connect the special terminals on the power supply to an audio oscillator and adjust to 400 cps output. Adjust the LF gain to obtain 1 volt ac superimposed on the 28 vdc supply. Adjust the AF gain control on the receiver for 200 milliwatts output indicated on the audio load. With the distortion analyzer connected to the audio load, the function switch to set level and the meter switch to 100% (+20 db), adjust the input control to 0 db as indicated on the distortion analyzer meter. Remove the 400 cps modulation by turning the LF range control to OFF. Rotate the distortion analyzer switch counterclockwise adding 10 db for each detent passed until a meter indication is reached. Subtract the indication from the sum of the 10 db steps if the meter reads above 0 db and add the reading if below 0 db.

4.8.18 Noise limiter (see 3.10.17).- With the receiver controls set as in 4.7, the signal generator shall be tuned for maximum receiver diode load voltage at 1.5 mc, 50 microvolts output and 400 cps modulated at 30%. The receiver function switch shall be set to noise limiter and AF gain adjusted to obtain 10 mw audio output. Connect an oscilloscope to the audio output load terminals. Adjust the oscilloscope to obtain the waveform of the 400 cps modulation. Increase the modulation percentage by turning the LF gain on the signal generator clockwise until clipping of the 400 cps audio signal is noted (flattening of the wave peaks). Record the percent modulation at which clipping is first observed.

4.8.19 IF output characteristics test (see 3.10.18).-Set controls as in 4.7 and tune the receiver to 7.5 megacycles. Connect a standard input signal of 3 microvolts through a 56- micromicrofarad capacitor to the ANT terminal. Measure the IF output voltage with an rf voltmeter at all settings of the BANDWIDTH switch and AGC switch ON and OFF. Record these outputs. Raise the input signal to 5 microvolts and record the resulting output voltages as above. Remove the input signal, then measure and record the IF noise levels.

4.8.20 Break-in relays (see 3.10.19).- With the receiver controls set as in 4.7, the signal generator and receiver tuned to 30.05 mc, set the signal generator attenuator to 10 uv with 400 cps at 30% modulation. Note the diode load voltage as indicated on the DC VTVM connected to J-615. Adjust AF GAIN to obtain 400 cps tone from speaker.

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Throw the Break-in Relay and note that 400 cps tone disappears. Advance generator attenuator until the same diode load voltage is obtained. Record this setting as antenna grounding relay attenuation. If tone is absent audio disabling relay is functioning.

4.8.21 Tuning knob torque test (see 3.10.20).- A torque of 72 inch-pounds shall be applied, alternately 250 times in each direction for a total of 500 times.

4.8.22 Backlash (see 3.10.21).- A dial indicator shall replace the Kilocycle knob. Turn the dial indicator clockwise for a given reading at a frequency counter. Note dial reading. Turn dial indicator clockwise a full turn, then counterclockwise for some reading on the frequency counter. Note dial reading. Fasten shaft of VFO so that it will not rotate. Turn dial indicator knob clockwise without turning VFO shaft; note dial indicator reading. Also note dial indicator reading for ccw rotation of VFO shaft. Convert differences in dial indicator readings to cps to determine compliance with 3.10.21.

4.8.23 Megacycle dial calibration error (see 3.10.22).- The BFO switch shall be adjusted to the ON position, and the AGC switch set to CAL. Tune the receiver to the 100 kc calibrator signal at 500 kc to produce an audio frequency beat note. Without further adjustment of the BFO or of the kilocycle tuning control, rotate the megacycle tuning control through its 32 steps. At each step, the audio output frequency of the receiver shall be measured and recorded. The initial adjustment at 500 kc shall be detuned from zero beat so that at each megocycle step, the receiver tuning will remain on the same side of zero beat.

4.8.24 Frequency identification accuracy (see 3.10.23).- A signal generator shall be coupled to the receiver antenna input, and a one megacycle interval in the receiver tuning range selected. The indicating counter shall be zero adjusted at the nearest multiple of 100 kc by adjusting the BFO so that zero-beat occurs when the counter reads exactly on the 100 kc multiple. Backlash error may be eliminated by tuning in one direction to the frequency at which the error is measured. Error shall be measured with a Berkley Eput Meter Model 554M, or equal. Test frequencies, as shown on the KILOCYCLES counter shall begin with 50 kc, and shall progress in 50 kc increments to 950 kc.

4.8.25 Zero adjust mechanism (see 3.10.24).- The test shall be conducted at 500 kc and at 32 mc. A stable external signal source shall be provided. The receiver BFO shall be turned ON and adjusted for a zero-beat. Push in and adjust the DIAL ZERO control for + 6 kc and -6 kc variations in dial reading at each test frequency. There shall be no change in receiver tuning.

4.8.26 100 kc Crystal Calibrator (see 3.10.25).-

4.8.26.1 Crystal calibrator output (see 3.10.25.1).- The signal-plus-noise to noise ratios of the calibrator signals having the lowest output shall be measured by observing and recording the audio power output with the BFO switch ON and the AGC switch in the CAL position, and again measuring the audio power output with the AGC switch in the ON position. When the AGC switch is in the CAL position, the receiver BFO shall be adjusted for 1000 cps.

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4.8.26.2 Crystal calibrator cycling (see 3.10.25.2).- The receiver shall be tuned to the calibrator signal at 30.0 mc, BFO switch ON. The difference between maximum frequency variations in the audio output shall be recorded.

4.8.27 Tracking of tuned circuits (see 3.10.25).- The function switch shall be set in the MGC position. Diode load voltage shall be observed, and signal input shall be below the blocking level as observed by change in diode load voltage with change in input signal. At each test frequency, mistracking shall be measured by adjusting the tuning slug for maximum diode load voltage. The increase in diode load voltage, in db, shall be the amount of mistracking. The slug shall be readjusted to its original position before proceeding to the next test frequency. Test frequencies (dial frequencies) shall be as follows:

5-1 mc	1-2 mc	2-4 mc	4-8 mc	8-16 mc	16-32 mc	9-18 mc variable IF	2-3 mc variable IF
.50 mc	1.00 mc	2.00 mc	4.00 mc	8.00 mc	16.00 mc	1.0 mc	1.999 mc
0.65 mc	1.25 mc	2.50 mc	5.00 mc	10.00 mc	20.00 mc	2.5 mc	1.75 mc
0.75 mc	1.50 mc	3.00 mc	6.00 mc	12.00 mc	24.00 mc	4.5 mc	1.50 mc
0.85 mc	1.75 mc	3.50 mc	7.00 mc	14.00 mc	28.00 mc	6.5 mc	1.25 mc
+0.00 mc	+1.00 mc	+3.00 mc	+7.00 mc	+15.00 mc	+31.00 mc	7.999 mc	1.00 mc

4.8.28 Desensitization (see 3.10.27).- The test frequency shall be 6.0 mc. The receiver service selector switch shall be in the NORMAL position, the AGC switch to ON. Desired and undesired signals shall be fed into the receiver under test from separate generators. The signal generator furnishing the desired signal shall first be connected to the receiver through a 56 uufd capacitor. Signal input to the receiver shall be 10 microvolts, modulated 30 percent at 400 cps. The signal generator and receiver shall be tuned to 6.0 mc and the receiver adjusted to produce a suitable audio output reference level. The 56 uufd capacitor shall then be replaced by a 10,000 ohm resistor and the generator output level raised to produce the reference audio level. The generator producing the undesired signal shall be connected to the receiver through a 56 uufd capacitor, and the unmodulated output of the generator adjusted progressively to levels of 0.1, 1.0, and 10.0 volts. At each level, the undesired signal shall be tuned toward the desired signal, from each side, until modulation due to the desired signal is reduced by 3 db. The frequency separation between the desired and undesired signals for each such set of conditions shall be measured and recorded.

4.8.29 Cross modulation (see 3.10.28).- The receiver service selector switch shall be set to the NORMAL position, and the AGC switch OFF. The IF BANDWIDTH switch shall be set to the 4 kc position. Desired and undesired signals shall be supplied to the receiver from separate signal generators. The generator furnishing the desired signal shall be connected to the receiver through a 56 uuf capacitor. Signal input to the receiver shall be 100 microvolts, 30 percent modulation at 400 cps. The receiver and signal generator shall be tuned to the test frequency and the receiver ANT TRIM control adjusted for peak response. The receiver of gain control shall be adjusted to

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the maximum clockwise position and the rf gain control adjusted to produce 100 milliwatts audio output. Then replace the 56 uufd capacitor with a 10,000 ohm resistor. The generator furnishing the undesired signal shall be connected to the receiver through a 56 uufd capacitor. The generator supplying the undesired signal shall be detuned and the output level reduced so as not to affect the desired signal. Without any adjustment of receiver controls, with the exception of ANT TRIM, the level of the desired signal shall be raised until 100 milliwatts of audio output is again attained. At this point, the AGC shall be turned on and the rf gain control set to the maximum clockwise position. Readjust the rf gain control for 100 milliwatts output. Modulation shall then be removed from the desired signal and applied to the undesired signal. The undesired signal, modulated with 400 cps at 30 percent, shall be set successively to levels of 0.01, 0.1, 1.0 and 10 volts. At each level, the undesired signal shall be tuned toward the desired, from each side, until modulation is produced on the desired signal 20 db below the reference audio level which is 1 mw. The frequency separation between the desired and undesired signals shall be measured and recorded for each set of conditions.

4.8.30 AGC time constant (see 3.10.29). - With the BFO switch in the ON position, signal input to the receiver shall be adjusted for a carrier level meter reading of 90. The signal input shall then be abruptly removed and the time required for the meter to drop to 30 shall be recorded as a measurement of the slow time constant. Repeat the test with the BFO in the OFF position. The time required for the carrier level meter to drop from 90 to 30 shall be approximately one-tenth of the time required when the BFO control is in the ON position.

4.8.31 Carrier level meter (see 3.10.30). - The reading of the carrier level meter shall be observed as the signal generator output is increased by 20 db steps from 5 to 500,000 microvolts. The AGC switch shall be in the ON position.

4.8.32 Blocking (see 3.10.31). - Test frequencies shall be 0.75, 1.5, 3.0, 6.0, 12.0, and 24.0 mc. The test shall be conducted at only one test frequency per unit, and the test frequencies shall be successively increased on the succeeding units tested. The generator shall provide a standard test signal (30 percent modulation at 400 cps) to the receiver input through a 56 uufd capacitor. A 5 microvolt signal shall be used to provide a reference level of 10 milliwatts at the audio output. Audio output shall be observed as the input level is slowly raised to 0.2 volt.

4.8.33 VFO stability vs temperature (see 3.10.32). - The VFO assembly shall be placed in a temperature chamber. Test frequencies shall be 2.5, 3.0, and 3.4 mc. The initial temperature shall be +50°F (+10°C). The temperature shall be increased in steps of 18°F (10°C), and temperature stabilization shall be reached at each step before recording the VFO output frequency. The maximum frequency change shall be considered the difference between the highest and lowest of a frequency-temperature curve plotted from the data.

4.8.34 BFO frequency stability vs temperature (see 3.10.33). - Stability of the BFO shall be measured using the same procedure specified for the variable frequency oscillator. (See 4.8.33)

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4.8.35 BFO leakage at IF output (see 3.10.34).- The if BANDWIDTH switch shall be adjusted to the 4 kc position, and the BFO tuned for maximum leakage. Leakage shall be measured across a 50 ohm resistor at J-105.

4.8.36 20 hour frequency stability (see 3.10.35).- The receiver shall be tuned to a stable signal source of 8.5 mc and the BFO adjusted for a convenient audio output frequency, such that a positive frequency change at the VFO will result in a positive change in audio output frequency. The audio output frequency shall be measured and recorded three minutes after power is applied, and again after the second and the twentieth hours. Power shall not have been applied to the receiver for at least four hours prior to this test. The output frequency of the receiver shall be continuously recorded by automatic methods.

4.9 Temperature test (see 3.8.2).- The equipment shall be subjected to the temperature cycle shown on Standard MIL-STD-169. Performance shall be as follows:

Steps	Paragraph	Measurement	Performance
1, 3, 5, 8 and 10	4.8.11	Overall gain	Specified performance
	4.8.19	IF output	Specified performance
	4.8.14	Audio output	Specified performance
	4.8.24	Frequency Identification accuracy	Specified performance

4.10 Moisture-resistance test (see 3.8.3).- The equipment shall be tested as follows:

4.10.1 Test conditions.-

(a) Do not move equipment from the humidity chamber for measurements.

(b) Start measurements not more than 5 minutes after power is applied to the equipment. Complete measurements as rapidly as possible. Do not leave power on after measurements have been completed.

(c) Test sealed equipment with the seal intact except during last two cycles. Bring out leads for electrical measurements through a hole in the equipment enclosure and seal against entrance of moisture. Perform last two cycles with the chassis removed from its enclosure.

4.10.2 Test method.-

(a) Dry at 130° ±5°F for 24 hours.

(b) Condition at 77° ±5°F and 40 to 50 percent relative humility for 24 ours.

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(c) Take measurements as specified in 4.10.3, and readjust or realine as necessary to meet full specification requirements.

(d) Subject to continuous cycling for fifteen 48-hour cycles. Temperature, relative humidity, and period of time for each portion of the cycle shall conform to Standard MIL-STD-170. The measurements specified in 4.10.3 shall be made during the periods shown on the standard and shall comply with 4.10.3.

(e) After cycling has been completed, condition the equipment for 24 hours at 77° ±5°F and 40 to 60 percent relative humidity. Then adjust for optimum performance, using only those means provided by the equipment. No repair or replacement of parts shall be made. After adjustment, the equipment shall meet full specified performance for those measurements specified in 4.10.3.

4.10.3 Performance.- Measurements made during test and specified performance during cycling shall be as follows:

Paragraph	Measurement	Performance during cycling
4.8.11	Overall gain	Specified performance
4.8.19	IF output	Specified performance
4.8.14	Audio output	Specified performance
4.8.24	Frequency identification accuracy	Specified performance

4.10.4 Failure.- If any equipment fails to meet the performance specified in 4.10.3 during cycling, it shall be realined or readjusted once. If the equipment then fails to meet 4.10.3 or fails subsequently during cycling, it does not pass the test. In addition, if the equipment fails to meet full specification requirements after conditioning and adjustment, as specified in 4.10.2(e), it does not pass the test.

4.11 Elevation test (see 3.8.4).- The equipment, placed in its normal operating position in an altitude chamber, shall be tested as follows: (Note: To check temperature stabilization place a thermocouple on the largest internal mass centrally located within the equipment).

(a) With the equipment placed in the chamber, operating and stabilized at room temperature, overall gain (4.8.11), IF output (4.8.19), and audio output (4.8.14) shall be performed.

(b) Reduce the barometric pressure to 20.6 ± 0.1 inches of mercury (simulated 10,000 feet above sea level) and, after temperature and pressure stabilization, repeat the measurements specified in 4.11(a). The equipment shall meet specified performance for those measurements.

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(c) Remove the power from the equipment and further reduce the barometric pressure to 11.1 ± 0.1 inches of mercury (simulated 25,000 feet above sea level). Maintain this condition for 1 hour.

(d) Return the chamber to ambient pressure. Apply power to the equipment and, after stabilization, repeat the measurements specified in 4.11(a). The equipment shall meet specified performance for these measurements.

4.12 Orientation test (see 3.8.6). - The equipment shall be inclined for a minimum of 5 minutes in each plane (forward, backward, left, and right) to an angle of 20 +3 degrees. During inclination in each plane, the equipment shall meet specified performance for the following measurements:

Paragraphs	Measurement
4.8.11	Overall gain
4.8.19	IF output
4.8.14	Audio output
4.8.19	IF output

(f) Remove equipment from test chamber and allow it to cool t' room temperature. The equipment shall meet specified performance for the following measurements:

Paragraphs	Measurement
4.8.11	Overall gain
4.8.14	Audio output

(g) Open the equipment. There shall be no evidence of deterioration of either external or internal parts.

4.13 Shock test; ballistic (see 3.9.3). - The test shall be conducted on the "Shock Testing Machine for Light Weight Equipment" shown in Specification MIL-S-901. The equipment shall be secured in its normal operating position to the steel test plate by means of the same fasteners used for vehicular installation of the equipment. The test shall consist of a total of 9 blows: one each 1-foot blow, 3-foot blow, and 5-foot blow on the back, side, and top of the testplate. As an alternative to reorienting the test plate for the blows on the side of the plate, equivalent rotation of the equipment under test is permissible.

4.14 Shock test; bench-handling (see 3.9.3). - The chassis-and-front-panel assembly shall be removed fr.m its enclosure, as for servicing, and placed in a suitable position for servicing on a solid, 2-inch fir bench top. The test shall be performed as follows, in a manner simulating shock liable to occur during servicing:

(a) Tilt up the assembly through an angle of 30 degrees, using one edge of the assembly as an axis, and permit the assembly to drop back freely to the horizontal. Repeat, using other practicable edges of the same horizontal face as axes, for a total of four drops.

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(b) Repeat "(a)", with the assembly testing on other faces, until it has been dropped for a total of four times on each face on which the assembly might practically be placed during servicing.

4.15 Immersion test (see 3.8.5).- The equipment, as prepared for field transportation, shall be immersed to a minimum depth of 3 feet of fresh water for 2 hours. Immediately prior to immersion, the temperature of the equipment shall be 40°F, or more, above the temperature of the water. The tank in which the equipment is immersed shall be of sufficient capacity to maintain the water within ±2°F of its initial temperature, or the temperature of the water shall be maintained within those limits by other means. After completion of the 2-hour period of immersion, the equipment shall be removed from the water and wiped dry on exterior surfaces. When the equipment is opened, there shall be no evidence of leakage.

4.16 Vibration tests.- Internal vibration of the equipment shall be measured as follows, to determine conformance to 3.9.1.

(a) The equipment shall be fastened in its normal mounting position (shock mounts shall be blocked) on a vibration table that can be controlled within 10% of the specified amplitude.

(b) The vibration table shall provide approximate sinusoidal vibration. The equipment shall be vibrated successively in three mutual perpendicular directions that are respectively parallel to the edges of the equipment. In the frequency range from 10 to 55 cps, any frequency shall be used only long enough to determine whether a resonance exists. The frequency range from 10 to 55 shall be traversed by hand at 1 cycle per second increments with a constant total excursion of .0156".

(c) Resonances of components, structure, and subassemblies may be detected visually, by means of a Strobotac, as made by the General Radio Corporation, Cambridge, Massachusetts, or equal, and by energizing the equipment and detecting mechanical vibration through electrical output indications.

(d) If because of compactness or inaccessibility, it is not possible to determine the resonances completely by the above means, the equipment shall be broken down into logical subassemblies for this test. These subassemblies shall be secured to the vibration table in the same fashion that they are secured in the actual equipment.

4.17 Bounce test, (see 3.9.2).- The receiver on its mounting, shall be tested on the package tester, as made by the L.A.B. Corporation, Skaneateles, New York, Model 1000 SC or equal, as follows:

(a) Secure the equipment to the L.A.B., 48 by 48 inch, 305 pound, vehicular adapter plate, or equal, in the same manner as it would be mounted in a vehicle. The adapter plate shall be free of cracks and broken welds, and all unused holes except the 2-inch breather holes shall be plugged. No part of the mounted surface shall be less than 7 inches from the edge of the adapter plate.

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(b) Cover the tester bed with a panel of 1/2 inch plywood, with the grain parallel to the drive chain. Space sixpenny nails, with the heads below the surface, at 6 inch intervals around all four edges and at 3 inch intervals in a 6 inch square in the center.

(c) Place the adapter plate on the bed of the package tester. Limit the lateral motion, by wooden fences, to not more than 3 inches and not less than 1 inch. Additional barriers may be used to safeguard personnel, provided that the fore-and-aft motion of the adapter plate against the backstop is not restrained.

(d) Operate the package tester, shafts in phase, for a total of 3 hours at 283 ± 1 rpm. Turn the adapter plate 90 degrees at the end of each 45 minutes, each time in the same direction. Rest periods may be provided to permit cooling of the resilient mounts, provided that the testing time is not less than 3 hours. The receiver shall receive a standard teletype message, 850 cps frequency shift keyed during the bounce test.

4.18 Air-seal test (see 3.6).- The equipment shall be opened and closed again in such manner as to break and remake the seal. Immediately thereafter, the equipment as field transported shall be subjected to a vacuum of 1 pound per square inch (1 pound per square inch less than the atmospheric pressure surrounding the equipment) applied to the interior of the transit case or to the interior of the transit case or to the interior of the equipment inclosure when no transit case is provided. The vacuum then shall be valved-off and the interior pressure measured during the ensuing period of 1 minute. During this 1 minute period the decrease in vacuum shall not exceed .01 pound. The gage used for measurement of the vacuum shall be of such accuracy that a difference of .01 pound can be determined readily.

4.19 Visual and mechanical inspection.- Equipment shall be examined for the defects listed in Standard MIL-STD-252.

4.20 Inspection for interchangeability.- The dimensions listed below shall be gaged or measured to determine compliance with the physical interchangeability requirement of 3.5. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

(a) Width of chassis and gear plate, length of chassis and cam hub, crystal oscillator recess and 2 mounting holes, 2 chassis screws, 8 side mounting bracket holes, interunit contact, and location of kc and mc tuning shafts.

(b) Front mounting, rear area over screwheads, width over screwheads, 3 mounting holes and band-switch shaft from left side, and bottom of the chassis.

(c) Width and length of assembly, width of master oscillator compartment, 5 mounting holes, 1 contact pin, 1 switch shaft and clearance notch.

(d) Height of interunit contact pin.

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(e) Location of shaft, width of socket bracket cover, and maximum projection of wire, locating from the oscillator cover.

(f) Widlth of compartment, 2 rear mounting holes in floor plate and 4 outside mounting holes inside the plates.

(g) Interunit contact height.

(h) Width of compartment, 5 mounting holes in floor plate, and calibration oscillator shaft.

(i) Location of mounting surface of master oscillator saddle from side of frame and shaft centerline.

(j) Location of 2 dowel pins, 10 mounting screws, sides, top and bottom of frame, depth back of rear plate and projecting floor plate screws.

(k) Inside width, length, corner radii, depth, 2 dowel holes, and rim flatness.

(l) Location of 10 mounting holes from 2 dowel holes.

(m) Maximum width and maximum length.

4.21 Suppression of radio interference. - The receiver shall be tested in accordance with Specification MIL-I-11748.

4.22 Operational Inspection (see 3.10.10). - A talk-out test shall be performed, and the following checked at each of these frequencies: 1.520, 2.380, 3.420, 4.440, 7.040, 11.960, 15.040 and 19.950 mc.

(a) Satisfactory sidetone.

(b) Satisfactory voice quality.

(c) Correct tuning.

4.23 Acceptance inspection of preparation for delivery. - Acceptance inspection of preparation for delivery shall be in accordance with the applicable provisions of Specification MIL-R-12454.

5. PREPARATION FOR DELIVERY

5.1 Receiver, Radio R-329( )/URR shall be preserved, packaged and tested in accordance with the procedures specified in Specification MIL-R-12454.

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6. NOTES:

6.1 Intended use. - Receiver, Radio, R-392( )/URR is used alone or as part of Radio Set AN/GRC-19 ( ). It may be used for fixed or mobile service.

6.2 Ordering data. - Procurement document should specify the following:

(a) Title, number, and date of this specification and any amendment thereto.

(b) Level of packaging and level of packing required for shipment. (Level A, level B, or level C).

(c) Preproduction Inspection:

(1) Detail requirements covering preproduction packs including inspection to be performed thereon and descriptive details, plans, drawings, etc.

(d) Marking and shipping of samples.

(e) Place of final inspection.

(f) Technical literature required. (See 3.12).

(g) Quantity and running spare parts required. (See 3.12).

(h) Submission of the statement of treatment referenced in 3.4.9, as soon as possible after award of contract. This statement should be submitted to the contracting officer.

(i) If rough handling test or cyclic exposure test is required.

6.3 Color. - The color chip furnished by the procuring agency (3.4.4) will be color chip No. X-24087, and may be obtained upon request to Commanding General, U. S. Army Signal Supply Agency, 225 South Eighteenth Street, Philadelphia 3, Pennsylvania, ATTN: SIGSU-J4b.

6.4 Group C Inspection. - Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action. (See 4.5.3.2).

6.5 Location of operational inspection air-seal test. - It is desirable that the operational inspection (4.22) and the air-seal test (4.18) be performed at a location that will minimize handling (which might cause damage to the equipment) after this inspection is completed. Any preparation for shipment which would require breaking of the equipment seal should be accomplished prior to the air-seal test so that the seal may remain intact thereafter. It is recommended that the entire lot (including all previously inspected sample units) be sampled and inspected immediately prior to packaging.

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6.6 Nomenclature. - The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: R-392W/URR. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. (See 1.1).

6.7 Verification Inspection. - Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product, and will normally be identified by the categories listed below:

(a) Type A-- The total of that inspection set forth in the Quality Assurance Provisions of this specification or the contract. Included in this category is that amount of inspection referred to as normal and tightened inspection by Military Standard 105.

(b) Type B-- That Inspection set forth in the Quality Assurance Provisions of this specification or the contract reduced in amount under the reduced inspection provisions of Military Standard 105.

(c) Type C-- A reduced inspection procedure resulting in a material reduction in the amount of inspection set forth in the Quality Provisions of this specification. The amount of inspection is less than that provided for in type B and is based upon a consistently acceptable product resulting from a planned quality control system voluntarily employed by the contractor in the production of the product.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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